

Es. 1

$$m_a = 0 \quad M_e = \sigma_a^2 = \frac{1}{4} (9 + 1 + 1 + 9) = 5$$

$$R_a(f) = \sigma_a^2 \cdot T = 5T$$

$$\bar{R}_y(f) = R_a(f) \cdot |H(f)|^2$$

$$H(f) = \mathcal{F} \left[\text{sinc}^2 \frac{t}{T} \right] = T \text{triangle}(fT)$$

$$\bar{R}_y(f) = 5T^3 \text{triangle}^2(fT)$$

$$\bar{M}_y = \int_{-\infty}^{+\infty} \bar{R}_y(f) df = 5T^3 \cdot 2 \cdot \frac{1}{3T} = \frac{10}{3} T^2$$

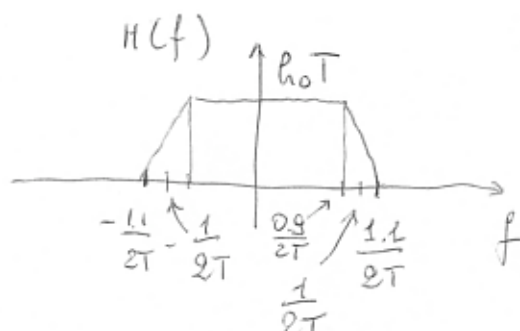
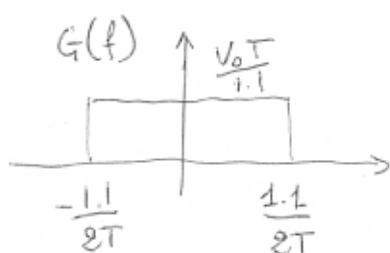
Es. 2

$$M_e = 2 \int_0^V \left(e - \frac{V}{2} \right)^2 \frac{a}{V^2} da = \frac{V^2}{12}$$

Es. 3

Con una banda di 8 MHz è possibile trasmettere fino a 8 Msimboli/s. Ogni simbolo trasmette $\log_2 16 = 4$ bit. Il bitrate massimo risulta 32 Mbit/s

Es. 4

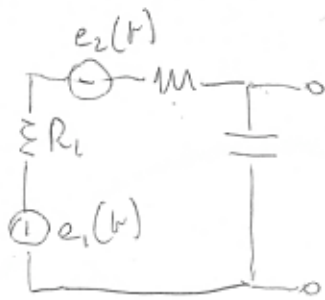


$$C(f) = \frac{V_0 T}{1.1} H(f) ; c(t) = \frac{V_0 h_0 T}{1.1} \text{sinc} \frac{t}{T} \text{sinc} \left(\frac{0.1}{T} t \right)$$

$$c(t) \text{ è di Nyquist con } c(0) = V'_0 = \frac{V_0 h_0 T}{1.1}$$

$$\sigma_m^2 = \int_{-\infty}^{+\infty} R_0 \cdot |H(f)|^2 df = R_0 \left(h_0^2 T^2 \cdot \frac{0.9}{T} + 2 h_0^2 T^2 \cdot \frac{0.2}{2T} \cdot \frac{1}{3} \right) ; P_e = Q \left(\frac{V'_0}{\sigma_m} \right)$$

E5.



$$V = (E_1 + E_2) \frac{\frac{1}{j\omega C}}{R_1 + R_2 + \frac{1}{j\omega C}}$$

$$R_w(f) = \frac{(2kT_1 R_1 + 2kT_2 R_2)}{1 + 4\pi^2 f^2 C^2 (R_1 + R_2)^2}$$